

Human Tooth Wear, Tooth Function and Cultural Variability^{1,2}

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ABSTRACT Among primitive peoples dental attrition appears to be a natural phenomenon. Often the degrees and kinds of tooth wear vary from population to population. This variability is possibly related to certain material aspects of culture such as diet, food preparation techniques and tool usage. In order to learn more about these relationships, extensive cross cultural comparisons must be made.

This paper reports on a study of dental attrition among skeletal remains of North American Indians from three areas: California, the Southwest and the Valley of Mexico. A method of comparing worn teeth of these populations was devised so several characteristics of the teeth and supporting bone could be examined by population. This study showed significant differences in type and degree of wear among the three groups as well as differences between sexes within each population.

A positive correlation between tooth wear and cultural factors was found. Dietary specialization and division of labor appear to be responsible for the degree and type of wear found in this sample. Further studies of this type are planned to expand the sample size and, if the new data support these correlations, valuable information about human-environmental relationships can be gained.

It has been recognized for some time now that many of the functions of the primate dentition have gradually shifted to the hands (DuBrul, '64:18). Food collecting and feeding-foraging activities in general have caused certain major alterations to occur in the primate face and dentition during evolution. The degree of change and the extent of environmental influence on relative sizes of the primate jaw mass and teeth are perhaps some of the most important questions involved in a study of the earliest fossil hominids. A change in feeding habits or a developing reliance on tools are believed to be prime causes for the reduction in tooth size in family Hominidae (Simons and Ettel, '70:82; Washburn, '59:31). The increased use of mechanical methods in food preparation gradually shifted food reduction from inside the mouth to the outside, and this was a major factor in modifying the

lower part of the face (Campbell, '39, part 11:53).

Even after the time of the reduction of the canines and the shift of the tool-defense functions of the dentition to the hand-held implement the teeth played an important role in the survival of the individual (Dahlberg, '63:172). The evidence for survival value of a large oro-facial complex is provided by the heavily worn teeth of many prehistoric skeletal populations. The heavy dental attrition among prehistoric groups and even in many living primitive populations suggest that the

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wearing away of tooth surface is a natural biological process (see Goose, '63:128; Klatsky, '39:73, Sicher, '53:310, and Sognnaes, '63:93-94). From the Neolithic to the present day there has been a wide range of differences, both attritional and morphological, in the dentition of various human populations. While some kinds of dental wear may indicate that the teeth had been used for purposes other than eating, few studies of tooth wear among primitive people have been made. In addition, to my knowledge, no systematic attempt has been made to correlate the degree of pattern of attrition with tooth usage, either dietary or non-dietary.

The planes formed on the occlusal surfaces of the teeth by the wear patterns need to be thoroughly studied. Not only should this study reveal details about tooth usage but, in addition, it will yield information which will aid in the description of the motions and forces that had been applied to the dental arches. In effect, chewing activities that were engaged in by the individual during life are partially preserved in the skeletal record by the wear patterns. If the distribution of forces and motions over the mandible during mastication is understood then evolutionary development of the dental apparatus of the family *Hominidae* can be more effectively described. This description could be in terms of the selective pressures on the entire oral anatomy and especially the change in certain of these pressures as the "tool" functions shifted from the mouth to the hand.

It is probable that the kinds and degrees of tooth wear are related to the culture of the population, for certain studies have shown that dental attrition decreased with increasing urbanization (Davies and Pedersen, '55:43; Brothwell, '63a:279; Greene et al., '67). Brothwell noted that even in dry, sandy environments, where extensive dentine exposure commonly occurs, there is a decrease in this tooth destruction which is correlated with an increase in cultural development. In a Mesolithic population from the Sudan, Greene, et al., ('67:52) found a mixture of types of occlusions which they attributed to the transitional nature of the cultural phase. In addition, Anderson, ('65:49649-7), in a survey of the skeletal

material found in Tehuacan, Mexico, stated that there were relationships between changing dental wear patterns and the economy. This covariation of wear patterns and attrition with culture was emphasized when Moorrees, ('57:133) noted the limited usefulness in estimating age by comparing degrees of attrition, and Brothwell, ('63b:61) also listed similar difficulties in estimating age by use of worn teeth. In addition, Campbell observed a marked change in the degree of attrition between certain Australian Aborigines which he believed was due to the fact that some of them were as he put it, "... living under civilized conditions, ('38:128)."

Moorrees, ('57:129) defined dental attrition as the frictional wear of the teeth, and abrasion was described as the result of abrasive material food. Much of the loss of the occlusal surface is a result of a combination of several factors and these include the frictional and abrasive causes listed above, but the use of teeth as tools can account for a large amount of attrition in certain populations. The various sources of tooth wear will be considered and the term attrition will be used to describe all forms of enamel and dentine reduction.

Taking into account all of these factors of tooth usage and anatomical variation, the argument presented here is that a meaningful correlation can be made between human dental attrition patterns and the way in which the teeth were used during the individual's life. Typically, this usage is a function of the kind of food in the diet, and of the methods of food preparation as well as certain manipulatory functions for which the teeth are frequently used. With the aid of such correlations, the examination of skeletal remains gained by archaeological excavations can yield many valuable data relating to the former activities of the individuals. Such methods can be expected to provide a feedback of information between osteological and archaeological materials which can lead to a broader understanding of the technology and environmental conditions involved. Some of the information thus gained might demonstrate a sexual division of labor, variation in diet, the effects of dental pathologies, and the

influences of various craft specializations. Viewed in this way, cross-population comparisons of dental wear then should show significant differences that are related to the cultures under study.

However, one of the major handicaps to any cross-cultural comparison of dental attrition has been the lack of a suitable method for recording various features of tooth wear and associated alveolar bone lesions. Formerly, methods used by Broca and Hrdlička established a graded series for human molar wear and ignored the rest of the dental arch. Recently dental studies have used new categories for recording wear (see Brothwell, '63b; Greene, et al., '67). But all of these studies, the earlier analyses of human teeth as well as more recent reports, concerned themselves only with the degree of wear and so were based on gradients of dentine exposure. No record was made of the form of the occlusal surfaces that were the result of wear. Since the shape of the wear planes along these surfaces varies from population to population then this information must be described in some way so cross-cultural comparisons can be made. The following discussion outlines a method of recording these occlusal surface differences that will permit comparisons of unlimited numbers of populations or of individuals.

Tooth wear evaluation method

There are several problems involved in recording tooth wear. Though the degree of wear can be indicated; by using a series of numerals to represent the wear, no single series can be applied in the same way to record wear on all types of teeth, i.e., incisors, canines, premolars and molars. Recognizing this difficulty, a method was used that took into account the degree and kinds of wear to which each type of tooth is subject. The canines and incisors were evaluated together and the various degrees of wear were assigned categories (see fig. 1) on the basis of the criteria on dentine exposure. The direction of wear (fig. 2) and occlusal surface form (fig. 3) were evaluated separately by assigning numbers ranging from 1 to 8 and appropriate for recording in single column on electronic data-processing

cards. The criteria are discussed in greater detail below where the molars are considered.

Setting up recording procedures for the molars was relatively easy because of their multiple-cusp form. Therefore, the criteria for attritional evaluation were more precise and less arbitrary than those for the canines and incisors. The abrasion of the cusps and the appearance of dentine patches were used as a basis for evaluation as illustrated in figure 1. This system follows quite closely that of Brothwell's with an assist from Murphy's precise classification, ('59).

The premolars proved to be the most difficult to record and their wear ratings were usually much more subjective than either of the other kinds of teeth. However, using the same type of diagramming technique applied to the molars proved to be a great aid. Additionally, the premolar tooth proved to be the most variable in its wear patterns and degrees; at least in the samples used here.

This rating technique, which utilized wear diagrams for different teeth, provides an improved method over other systems, but it is essential to consider the wear planes in addition to the degree. In order to record the wear planes accurately, consideration must be given to all of the possibilities or variants in form and angle that the occlusal surface assumes with wear.

The type of wear must be described in two parts; first, the direction of the occlusal surface must be considered as in the case of obliquely worn molars; secondly, the surface form itself must be noted. Surface forms can be flat and smooth, or concave, giving the appearance of having been scooped out or "cupped," (see fig. 3). After an inspection of several hundred sets of dentition from California and Southwestern Indian remains it was found that there are several forms of worn occlusal surfaces for each kind of tooth. Certain of these types of wear appear more frequently and these types were used to establish the method for evaluation. These wear types can be recorded by a two place number system in which each place uses any one of nine digits.

The direction of the occlusal surface will be considered first. Each of the pos-

Category of Wear	Incisor and Canine	Premolar	Molars
1	Unworn.	Unworn.	Unworn.
2	Wear facets minimal in size.	Wear facets, no observable dentine.	Wear facets, no observable dentine. <i>exposed.</i>
3	Cusp pattern obliterated, small dentine patches may be present.	Cusp pattern partially or completely obliterated. Small dentine patches.	Cusp pattern partially or completely obliterated. Small dentine patches. <i>exposed on fewer than 3 cusps.</i>
4	Dentine patch (Minimal).	Two or more dentine patches, one of large size.	Three or more small dentine patches.
5	Dentine patch (Extensive).	Two or more dentine patches, secondary dentine may be slight.	Three or more large dentine patches, secondary dentine, none to slight.
6	Secondary dentine (Moderate to Extensive).	Entire tooth still surrounded by enamel, secondary dentine moderate to heavy.	Secondary dentine moderate to extensive, entire tooth completely surrounded by enamel.
7	Crown (enamel) worn away on at least one side, extensive secondary dentine.	Crown (enamel) worn away, on at least one side, extensive secondary dentine.	Crown (enamel) worn away on at least one side, extensive secondary dentine.
8	Roots functioning in occlusal surface.	Roots functioning in occlusal surface.	Roots functioning in occlusal surface.

Fig. 1 Degree of wear.

sible directions is listed and assigned a number. The direction, in the case of an oblique surface, is considered to run from the highest point to the lowest point on the occlusal surface (see fig. 2). The categories and their representative wear directions are listed below.

Category

- 1 natural form
- 2 oblique (buccal-lingual direction)

- 3 oblique (lingual-buccal direction)
- 4 oblique (mesial-distal direction)
- 5 oblique (distal-mesial direction)
- 6 Horizontal (perpendicular to the long axis of the tooth)
- 7 Rounded (buccal-lingual direction)
- 8 Rounded (mesial-distal direction)

Once the direction of the occlusal plane is recorded then the form of the surface

Scrub marks

must be considered. Since there is a difference in hardness between the enamel and dentine, the dentine, when exposed, will wear faster than the surrounding enamel of the crown. This differential density frequently results in the development of a concave or "cupped" surface. But there are at least two variations which must be considered and provided for in the category. These are the crowns of certain molars which are only partially cupped; that is, in cases where the dentine has been exposed in only a part of the tooth. In these instances approximately one-half of the surface will be cupped. This feature appears frequently in the case of the lingual cusps of the maxillary first molars. In addition, special tooth functions often will cause rather unique patterns of wear, and these patterns sometimes appear as "notched" or rounded, especially in the case of the incisors. The rating for the occlusal surface form will then appear as listed below. (See fig. 3).

Occlusal surface form

Category

- 1 Natural form
- 2 Flat surface
- 3 One-half of surface cupped
- 4 Entire surface cupped
- 5 Notched
- 6 Rounded

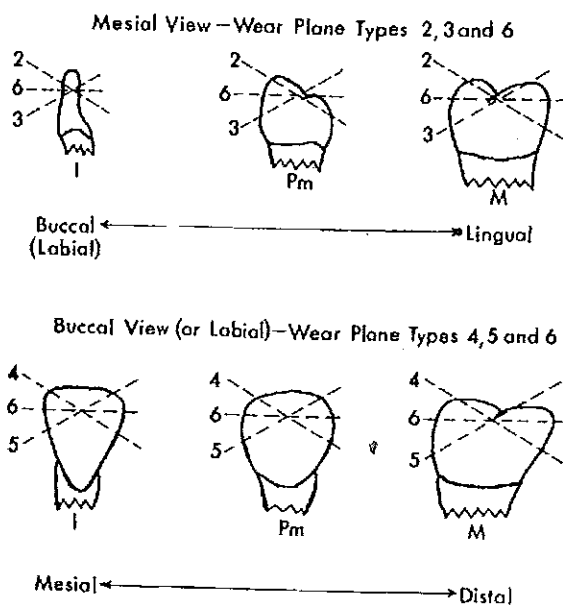


Fig. 2 Wear types—direction of surface wear.

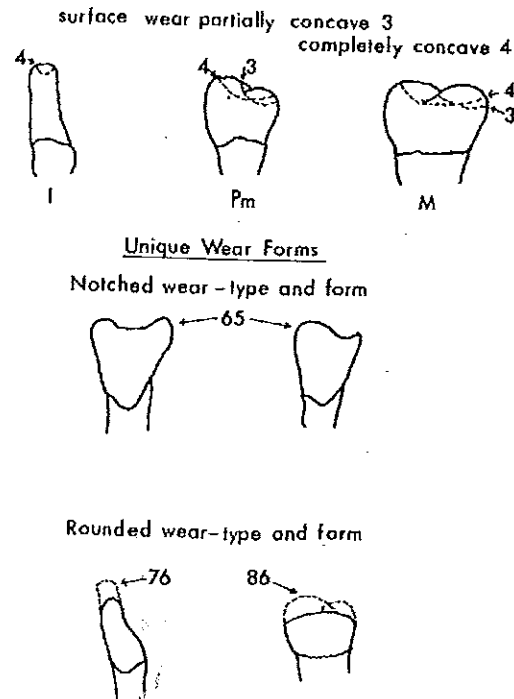


Fig. 3 Wear types—form of surface wear.

Certain of these forms listed above will (figs. 2, 3) more commonly appear on particular teeth. The molars will be "cupped" more frequently when heavily worn, while the incisors and canines are usually rounded when they reach a certain degree of wear. Adding together the ratings for the degree of wear and the types of wear, three digits can be used to describe the tooth in various stages of attrition. The application of these numbers allows for many different combinations and offers flexibility while still maintaining a considerable amount of accuracy in assessing dental attrition. The accuracy is increased further by providing different criteria of wear degree for each tooth. An incisor that has lost less than one-fifth of its crown through wear and has a minimal dentine patch would be rated as a 4. If its occlusal surface is flat in a horizontal direction it would be a 62. However, in order to fully describe the tooth a combination of all three of these digits would code the tooth as 624. If a molar was described as a 624 then the degree or number 4 would indicate that it had three or more small dentine patches. But the type of wear would be the same as the incisor or 62 which indicates a horizontal flat occlusal surface.

Once the description of the worn tooth has been accomplished the next step must be to record this description in a way which allows for the most efficient manipulation of the data. The standard 80 column card provided a ready system. Because of the space requirements, four cards were used to record the teeth from a single skull, two cards for each dental arch.

The individual teeth were grouped on each card according to their functional relationships. Therefore, the maxillary incisors and canines were placed on one card while the molars were recorded on the other. It was decided to place the first premolar with the incisor-canine group and the second premolar with the molars.

More is involved in the consideration of each tooth than merely an appraisal of its wear patterns, and certain of these data were included on each card: descriptions of the condition of the alveolar bone, presence or absence of dental caries, and tooth eruption. Age, sex, arch widths, and kind of occlusion were also recorded. The inclusion of this information in the data record for each skeleton made it possible for extensive cross-population comparisons to be made as described below.

The sample

Dental attrition information was collected from several populations in order to examine the possibility that patterns and degrees of tooth wear are related to diet and tooth use. Several North American Indian skeletal collections were carefully examined and those individual skulls with fairly complete dentitions were selected. This reduced the sample size but a careful selection was necessary in order to obtain as complete a record as possible in order to evaluate the method described above. The skeletal collections were from the following areas.

- I. California
- II. Southwest
 - a. Gran Quivira, New Mexico
 - b. Cibecue, Arizona (The Grasshopper site)
- III. Valley of Mexico
 - a. Tlatilco
 - b. Cuicuilco

A total of 317 sets of dentition from the three groups were studied. All were evaluated and photographed and 162 were recorded on data processing cards. The following is a brief summary of the cultural affinities of each group.

California—(39 dentitions)

This group consists of a collection of skeletal materials recovered during an excavation of a California Central Valley site, 4SJo17 located east of Stockton. The site is dated as "Middle Horizon" or 2,000-3,000 years B.P. on the basis of the artifactual assemblages and the burial styles (Heizer and Fenenga, '39). The absence of red ocher and arrow heads and the presence of numerous large pestles and harpoon points were the major items that identified the groups as "Middle Horizon."

The population had been non-agricultural but apparently subsisted well in an area which had an abundance of game and wild plant foods. Lance points and numerous animal bones indicate a quantity and variety of meat in the diet, principally elk, deer, duck and geese. The presence of harpoon points and many salmon bones suggest that fish were also an important part of the diet. The frequent occurrence of pestles indicate seeds of various grasses and most likely acorns were eaten in quantity.

Further elaboration on the environment of this group will have to await the monograph now in preparation by the archaeology staff at California State College at Long Beach but some impressions can be offered. Even though this population subsisted on a variety of foods, apparently of high nutritive value, they suffered from a high incidence of dental decay, and many pathological lesions were seen on the alveolar bone. Compared to coastal populations which I have studied, the Central Valley group was more subject to disease and had less tooth wear. It would fit between California coastal populations and Pueblo groups in terms of frequency of dental caries and degree of tooth wear.

The Southwest

A. *Gran Quivira, New Mexico—(41 dentitions).* This group is tentatively